Received: 2 March 2020 | Revised: 31 December 2020 | Accepted: 18 February 2021

DOI: 10.1111/1477-9552.12430

ORIGINAL ARTICLE

JAE Journal of Agricultural Economics

China-U.S. trade dispute and its impact on global agricultural markets, the U.S. economy, and greenhouse gas emissions

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Abstract

China is a major importer of agricultural products and we examine retaliatory tariffs imposed by China on U.S. pork, soybeans, corn, and wheat. We use an agricultural trade model to determine the impacts on agricultural commodity markets and combine our results with an input-output model to measure economic effects in the United States. In addition, we calculate global greenhouse gas (GHG) emissions from land-use change. The consequences of retaliatory tariffs are both trade destruction (lower overall trade) and trade diversion (trade diverted away from the U.S. and to other exporting countries). By the end of the projection period, total U.S. pork exports decline by 4.7%. Total soybean and wheat exports decrease by 31.2% and 0.5%, respectively whereas corn exports increases by 4.0%. Domestic U.S. retail pork prices decline by 0.8% and commodity farm prices decrease by 4.2%, 5.1%, and 15.8% for corn, wheat, and soybeans, respectively. The decline in foreign demand reduces U.S. production and welfare. Key among these impacts are a loss of nearly 15,400 jobs and a fall in labor income by \$1.35 billion due to a \$3.70 billion decline in national output at the beginning of the projection period. By the last year of the projection, the impacts grow to almost 30,000 fewer jobs and \$2.31 billion less labor income. The U.S. economy experiences a loss of slightly under \$6.80 billion in national output. Changes in trade due to the tariffs result in a reduction of GHG emissions from land-use change of up to 83.7 teragram (Tg) of CO_2-equivalent.

KEYWORDS
carbon emissions, commodity prices, crop exports, food prices, land allocation, tariffs

JEL CLASSIFICATION
C60; C54; C67; F18; Q02; Q15; Q17; Q18

1 | INTRODUCTION

China is a significant net importer of several agricultural commodities for which the United States is a major exporter—for example, pork, soybeans, corn and wheat. Thus, policies that curtail the demand for exports from the United States to China can potentially have large negative domestic economic impacts for both countries. In May 2018, the United States imposed a 25% tariff on imported steel and a 10% tariff on imported aluminium from all countries including China. Several countries were excluded from the US tariffs once they agreed to a quota. The Chinese response was immediate, with tariffs on 128 items including agricultural products such as pork and ethanol (CRS, 2019a; Li et al., 2018; Marchant, 2018). Subsequently, the United States imposed import tariffs ranging from 10% to 25% on Chinese goods valued at \$250 billion. In return, China imposed up to 25% tariffs on \$60 billion worth of US goods including soybeans. To alleviate the short-run negative effects of the Chinese tariff on US farmers, the US government announced the introduction of three assistance programmes for 2018, which were expanded in 2019 (Giri et al., 2018).[1]

Although multiple countries have imposed trade restrictions on the United States and vice versa, we focus on the China–US conflict for three reasons. First, according to data from the US Department of Agriculture (USDA), China is by far the largest importer of pork and soybeans in the world, buying roughly 20% and 62%, respectively, of global exports in 2017. With a smaller market share (roughly 2%–3% of global imports in 2018/19), China is also an important player in the world trade of corn and wheat.[2] Given China's significance in trade, changes in trade policy have potentially significant impacts on global commodity markets. Second, the sheer size of the volumes exchanged among the countries in recent years imply potentially large trade reallocation across the world, with long-lasting effects on competitiveness and on long-term carbon emissions. Third, modelling trade effects from multiple countries jointly does not allow for the identification of each country's individual contribution to changes in production, prices and trade.

[1]US Department of Agriculture, Washington, DC: 'USDA announces details of assistance for farmers impacted by unjustified retaliation' (https://www.usda.gov/media/press-releases/2018/08/27/usda-announces-details-assistance-farmers-impacted-unjustifiedPress Release No. 0167.18) and 'USDA announces support for farmers impacted by unjustified retaliation and trade disruption' (https://www.usda.gov/media/press-releases/2019/05/23/usda-announces-support-farmers-impacted-unjustified-retaliation-andPress Release No. 0078.19).

[2]Calculated from trade data obtained from USDA Production, Supply, and Distribution Database (PSD Online).

Brazil has been dominant in the global soybean market as a result of its strong growth in productivity, displacing the United States in the Chinese market (Yao et al., 2018). As part of our analysis, we estimate the greenhouse gas (GHG) emissions associated with the trade conflict because Brazil is an important agricultural producer, represents a significant biomass and soil carbon pool, and is expected to fill some of the trade space that opens up as China looks for alternative suppliers. Significant changes in agricultural markets affecting Brazil (but also other countries) can potentially have important GHG implications. The effects of trade restrictions on GHG emissions is not clear prior to trade modelling. Import tariffs interfere with trade allocating production to countries with a comparative advantage. This does not necessarily mean that countries with the lowest GHG emissions produce agricultural products in the absence of trade restrictions. Beyond a priori ambiguous change in the direction of net GHG emissions, some authors point to strong expansions in deforestation levels in Brazil (Faria, 2016; Fuchs et al., 2019; Pendrilla et al., 2019; Schmitz et al., 2015). They project potentially large levels of deforestation and increases of GHG emissions from Brazil as a result of the trade dispute being analysed here. More muted responses (and more in line with the results of this study) are obtained by Richards et al., (2020), using a model that allows for interactions across commodities and trade reallocation. A net decrease in emissions are found in Liu et al., (2020) with a large decrease in the United States and China but an increase in other countries. However, their emissions sources extend beyond GHG emissions from land use change and agriculture.

Studies have emphasised the interaction between trade and land use change from various angles. Meyfroidt et al., (2010) assess whether countries, which have shifted from net deforestation to net reforestation, have transferred forest loss to other countries. Results show that 74% of the reforestation area has been shifted to other countries through trade in crops and forest products. This highlights the interactions between changes in trade patterns (as presented in this paper) and land use change. Meyfroidt et al., (2010) focus on differences in forest areas between countries and not on differences in carbon through variations in soil and biomass characteristics. In this paper, we extend the analysis by Meyfroidt et al., (2010) by also including carbon emissions from land use change.

Trade liberalisation—that is, the absence of restrictions—allocates crops and livestock production to countries that have a comparative advantage and are not necessarily efficient from a GHG perspective. Schmitz et al., (2012) show that in the case of trade liberalisation, GHG emissions are higher from deforestation in Latin America and from livestock production in China. Latin America has a comparative advantage in crop production whereas China specialises in livestock production and hence, increasing non-CO_2 emissions. This shift in trade also has implications on biodiversity associated with land use change (Chaudhary & Kastner, 2016). Richards et al., (2020) assess one of the issues for one of the countries included in this paper. The authors analyse the impacts of land use change and GHG emissions in Brazil as a result of a tariff on US soybeans by China. They highlight the offsetting effects of pasture in reducing total land use change and GHG emissions from an expansion in cropland.

Estimates of (micro) trade elasticities by Grant et al., (2018) indicate countries' rapid change of trading partners in response to changes in the relative tariff levels. Some of the economic effects are already evident, with China shifting some of its soybean imports away from the United States towards Brazil, and soybeans from Brazil commanding increasing price premiums relative to the United States for a period of time (Zhou et al., 2018). Trade barriers between the United States and China can also incentivise the development of improved and lower cost infrastructure in Brazil to potentiate its commercial advantage. Sabala and Devadoss (2019) analyse the trade effects of retaliatory Chinese tariffs and find that Brazil experiences large welfare gains due to higher exports and that the United States as well as China incur welfare losses. Trade reallocations can also increase or decrease the levels of

GHG emissions associated with agricultural production and land use change. All of these developments can have long-lasting consequences for US producers, global agricultural markets and the environment.

Tariffs disrupt trade and consumption patterns leading to lower domestic prices and production for countries facing the tariffs and also result in lower overall welfare and economic activity when compared to a situation without tariffs. The main objective of this study is to analyse the impacts of retaliatory tariffs by China on US and global agricultural markets, on US economic activity, and on GHG emissions. We quantify these impacts by using a combination of (1) a well-established agricultural trade model, (2) an input-output model, and (3) a GHG model. The CARD model is the agricultural modelling system developed by the Center for Agricultural and Rural Development (CARD) at Iowa State University. The input-output model, IMPLAN (IMpact Analysis for PLANing), is a widely used framework to evaluate economic changes and policy. The GHG model is designed to quantify changes in GHG emission associated with agricultural land use change and is used in conjunction with the CARD model (Carriquiry et al., 2020; Dumortier et al., 2021).

The CARD model provides the impact on US and global agricultural commodities in terms of supply, utilisation and prices. The results from the CARD model are used as input to an input-output model of the US economy, that is, the IMPLAN model. The input-output model provides industry-level impacts in terms of changes in employment, labour income, industrial output and value added. Both models capture the linkages between sectors, which is important given the interrelations of supply and demand for agricultural products. Factors that disrupt the market for one commodity are likely to affect other sectors of the agricultural markets (Yao et al., 2018). The output from the CARD model is used in the GHG model to estimate emissions from carbon stock changes due to land use change.

Impacts of Chinese tariffs on US soybeans, wheat, corn, sorghum and beef (but not pork) on bilateral and global trade, production and welfare are analysed by Taheripour and Tyner (2018) using a computable general equilibrium model (GTAP-BIO) developed within the Global Trade Analysis Project (GTAP) at Purdue University. Sabala and Devadoss (2019) develop a spatial equilibrium trade model to analyse trade reallocations and welfare effects of a retaliatory tariff on soybeans alone. Using a partial equilibrium model, Zheng et al., (2018) consider Chinese retaliatory tariffs on soybeans, cotton, sorghum and pork. We extend previous analyses by including corn and wheat (though we do not deal with sorghum and cotton), by computing a different set of production and trade impacts, as well as indicators of economic effects (e.g., changes in value added and in the domestic labour market), and by analysing environmental implications of the trade dispute.

2 | MODELLING APPROACH

The CARD model generates 10-year price, supply and utilisation projections for global agricultural markets covering crops and livestock for 58 countries and regions. The CARD model has been used in various studies to analyse the impact of policies and shocks on agricultural markets and has been described elsewhere (Carriquiry et al., 2020; Dumortier et al., 2011; Hayes et al., 2009; Searchinger et al., 2008). In our discussion of the CARD model, we focus on changes we make to identify the effects of the retaliatory Chinese tariffs. The output of the CARD model is coupled with the input-output model IMPLAN and a GHG model (the GHG model). The latter has been used with regard to biofuel and livestock policies to quantify GHG emissions from changes in land allocations (Carriquiry et al., 2020; Dumortier et al., 2011, 2012). These publications also describe the evolution of the CARD model since the Searchinger et al., (2008) paper to better model global agricultural markets and land use. In particular, the model has been updated and improved over time to incorporate the feedback received from

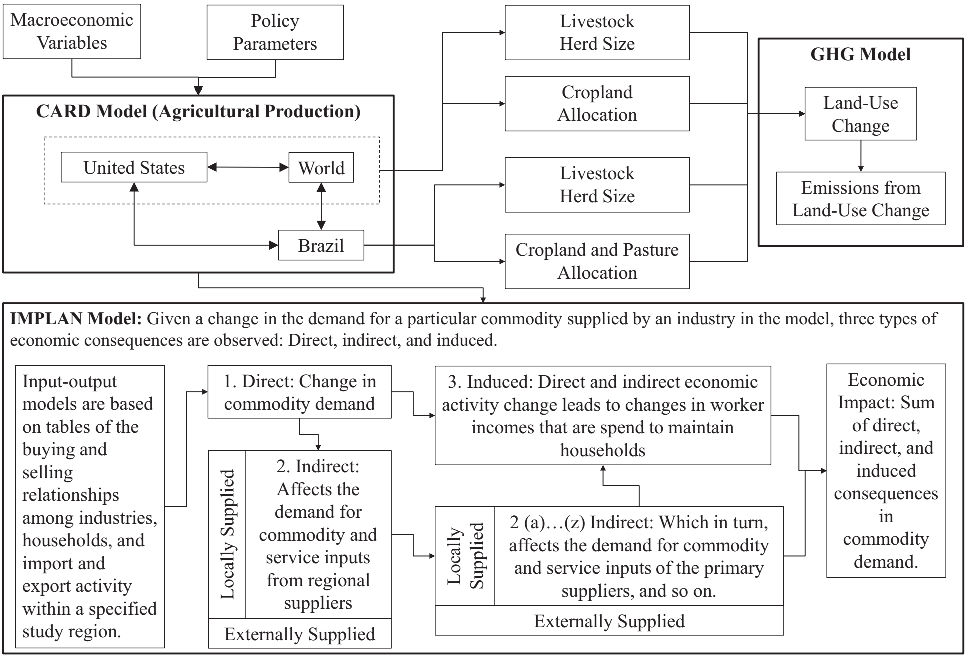

FIGURE 1 Model structure. The results from the CARD model are used in the IMPLAN model to calculate the effects of trade restrictions on the US economy. Global area and livestock as well as Brazilian pasture results are used to calculate GHG emissions associated with the trade restrictions

the Searchinger et al., (2008) paper. The most important changes are the implementation of a yield-price relationship, regional (as opposed to national) modelling of agricultural production in Brazil, and the absence of deforestation for agricultural expansion in the United States. As shown in Dumortier et al., (2011), especially in terms of the yield-price relationship, that is, higher crop prices resulting in above trend yields due to improved management practices (among other factors), has significant effects on land use and carbon estimates. Figure 1 illustrates the interaction of the models, and the following subsection briefly outlines their main features.

2.1 | CARD model

The CARD model is a system of econometric, partial equilibrium, non-spatial models for global agriculture. The model covers all major temperate crops, sugar, biofuels, dairy, livestock and meat products for all major producing and consuming countries. Extensive market linkages exist in the modelling system, which reflect derived demand for feed in livestock and dairy sectors, competition for land in production, and consumer substitution possibilities for close substitutes such as vegetable oils and meat types. For example, the demand for soybean oil and soybean meal takes into account prices of other oils and meals and vice versa, that is, the cross-price effects are included in the model.

The CARD modelling system is run to generate a baseline, which establishes 10-year projections by country and by crop for production, consumption, prices and trade. Within each model are behavioural equations for the supply and demand of the modelled commodities by country and these equations are the key drivers in the system. Each commodity model in the system is solved iteratively until an equilibrium is reached in each market by equating supply and demand year by year in the projection period. In general, a market-clearing price for a commodity is determined to equate supply and demand. Given that the markets are interlinked, changes in one commodity market impacts other markets in the modelling system. A set of exogenous macroeconomic variables as well as current agricultural and trade policies for each commodity in a country, which affect the supply and demand decisions, also drive the projections. Once the baseline is established, certain components of the modelling system are modified to simulate a specific scenario and the system is rerun to obtain a new market equilibrium for all commodity markets in the model. For our particular scenario, where we impose retaliatory tariffs, it is important to note that Chinese imports of some of the commodities affected by the tariffs are increasing over time in the baseline. This will likely affect the patterns of change of several relevant variables (e.g., US prices and exports) over time.

Prior to this study, the CARD model was non-spatial in specification and therefore bilateral trade between countries was not tracked directly. The non-spatial modelling version solved for world reference prices using a single world market-clearing mechanism that equalises net imports and net exports without regard to country sources of imports and destinations of exports. The US price for a commodity was considered the world reference price for many of the modelled commodities. A price transmission function transformed the US price to domestic prices by country/region to determine local demand and supply.

Because the retaliatory tariffs are imposed on the United States by China, the standard CARD model single-market clearing condition to determine a world price is modified to introduce a spatial dimension that captures differential price signals resulting from the retaliatory tariffs on US exports by China. Given the unprecedented nature of the trade dispute between the two countries, we adjust the model, which is initially non-spatial, to extract the trade between the United States and China from trade with other countries. We start with the CARD model used in Carriquiry et al., (2020) and implement the bilateral trade mechanism between the United States and China. We discuss these adjustments in some detail here because, to the best of our knowledge, partial equilibrium displacement models are not commonly used to evaluate bilateral trade disputes (Westhoff et al., 2019b).

Before we establish a baseline (business-as-usual or status quo scenario), we implement changes in the CARD modelling system and include additional equations to separate trade flows between the countries affected by tariffs (the United States and China) from trade with other commercial partners. In short, trade of products affected by the tariffs are separated from trade not subjected to the tariffs. We introduce the following additional specification to implement the necessary model changes. First, the historical shares of the United States in the imports of China are determined using 2012–2016 data provided by the Food and Agriculture Organization (FAO). An equation projecting the share of the United States in the Chinese import markets is added to allow for the disaggregation of Chinese imports by origin, that is, Chinese imports sourced from the United States and Chinese imports sourced from all other countries. The Chinese imports from the United States by commodity are determined as follows:

$$I_{CN,it}^{US} = s_{CN,it}^{US} \cdot I_{CN,it}^{T} \tag{1}$$

where $I_{CN,it}^{US}$ is China's imports from the United States of commodity i (i.e., pork, corn, soybeans, or wheat) in period t. $s_{CN,it}^{US}$ denotes the share of the United States in China's imports of commodity i in period t. The historical share of the United States in Chinese imports for each commodity is calculated as the ratio of China's imports from the United States to China's total imports from all countries (including the United States). For the baseline, the share in the projection period is the average of the last 5 years of history and is set exogenously at the level obtained from historical data. The baseline average historical shares for pork, soybeans, corn and wheat are 19%, 37%, 35% and 36%, respectively. The term $I_{CN,it}^{T}$ represents China's total imports of commodity i for

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all countries in period t. After China's imports from the United States are projected, then China's imports from other countries (excluding the United States) are given by:

$$I^{OC}_{CN,it} = I^{T}_{CN,it} - I^{US}_{CN,it} \tag{2}$$

where $I^{OC}_{CN,it}$ is China's imports from other countries (excluding the United States) for commodity i in period t. Second, because the United States is the world reference price in the case of pork, soybeans, corn and wheat, we introduce a new world price where only trade of the United States and China with other countries is accounted for together with the trade of all other countries. In this case, once China's imports from other countries by commodity are determined, these imports are entered into the new world market clearing solution to obtain a world price for each commodity. The market-clearing condition that determines the new world price is:

$$\sum_{j=1}^{n} NX_{j,it} = 0 \tag{3}$$

$NX_{j,it}$ denotes the net exports of country j of commodity i in period t. Note that the net exports (total exports minus total imports) can be positive for a net exporting country or negative for a net importing country. The equilibrium in the rest of the world excludes the intra-trade between the United States and China and solves for a market-clearing world price. Thus, Equation (3) includes China's net imports from other countries of commodity i in period t ($I^{OC}_{CN,it}$) but not China's imports from the United States ($I^{US}_{CN,it}$). Equivalently, Equation (3) includes the US net exports to other countries of commodity i in period t ($X^{OC}_{US,it}$) but not the US net exports to China of commodity i ($X^{CN}_{US,it}$).

Third, given the non-spatial specification of China's net trade with respect to the United States and other exporting countries, we ensure that domestic prices in China are determined by both the US price plus the tariff and the world price for each commodity. In the case of commodities where the model treats China as a large country (for pork, corn and wheat), the domestic price in China is solved endogenously and net trade is a behavioural equation that is a function of the weighted average of the US price plus the tariff and the world price. The weights are the exogenous shares of the United States in Chinese imports for each commodity imposed in the scenario. Where net trade is a residual (for soybeans), the domestic price evolution in China is driven by a price transmission that reflects price changes in the US and world markets. Given that the domestic price in China is transmitted from US prices for Chinese imports from the United States and from world prices for Chinese imports from other countries, net trade for this commodity in China is set as a residual to balance the domestic market. The residual net trade equation is given as:

$$NX_{CN,it} = Prod^{T}_{CN,it} + B_{CN,it} - U^{T}_{CN,it} - E_{CN,it} \tag{4}$$

where $NX_{CN,it}$ is Chinese net exports of commodity i in period t, $Prod^{T}_{CN,it}$ is Chinese total production of commodity i in period t, $B_{CN,it}$ are beginning stocks, $U^{T}_{CN,it}$ are total Chinese domestic use, and $E_{CN,it}$ are Chinese ending stocks. The price transmission equation for China by commodity is represented as:

$$P^{d}_{CN,it} = \gamma_{1i} + \gamma_{2i} \cdot \left[\lambda_i \cdot P^{m}_{US,it} \cdot (1 + \tau_i) + (1 - \lambda_i) \cdot P^{world}_{i}\right] \tag{5}$$

where $P^{d}_{CN,it}$ is the Chinese domestic price of commodity i in period t. The US market-clearing price of commodity i in period t is denoted by $P^{m}_{US,it}$ (see Equation 8). τ_i is the tariff in period t

(equal to 0 in the baseline). λ_i is the share of the United States in Chinese imports for commodity i in the scenario and P_i^{world} denotes the world price of commodity i.

Fourth, a mirror spatial dimension similar to China is added in the US model, that is, total US exports are disaggregated into exports to China and exports to all other countries for specific commodities. To balance the two markets, we impose by construction that US exports to China are equal to the Chinese imports from the United States for each commodity. That is:

$$X_{US,it}^{CN} = I_{CN,it}^{US} \tag{6}$$

where $X_{US,it}^{CN}$ is the US exports of commodity i to China in period t. US exports to all other countries are specified with a behavioural equation driven by the price in the domestic market relative to the price in the world. This equation is presented as follows:

$$X_{US,it}^{OC} = \alpha_i + \beta_i \cdot \frac{p_i^{world}}{p_i^{US}} \tag{7}$$

where $X_{US,it}^{OC}$ is the US exports of commodity i to other countries (excluding China) in period t. The world and domestic (US) price of commodity i are denoted p_i^{US} and p_i^{world}, respectively. The US country model has a market-clearing price ($P_{US,it}^{m}$) determination with trade to all countries in the following form:

$$Prod_{US,it}^{T} + I_{US,it}^{T} + B_{US,it} = U_{US,it}^{T} + X_{US,it}^{T} + E_{US,it} \tag{8}$$

where $Prod_{US,it}^{T}$ is US total production of commodity i in period t, $I_{US,it}^{T}$ are total US imports of commodity i in period t, $B_{US,it}$ are beginning stocks, $U_{US,it}^{T}$ are total US domestic use, and $E_{US,it}$ are US ending stocks. $X_{US,it}^{T}$ are total US exports and they are given by:

$$X_{US,it}^{T} = X_{US,it}^{CN} + X_{US,it}^{OC} \tag{9}$$

After the changes are incorporated, the CARD model is executed in a business-as-usual mode with status-quo policies (before the trade restrictions) to obtain a baseline. In a second step, the CARD model is executed again but now the retaliatory tariff scenario is implemented by imposing the tariff on each of the four commodities (pork, soybeans, corn and wheat), thus imposing border protection for Chinese imports from the United States. We use the same model for both runs, with the difference between the runs being the tariff rates and implied shares of US imports of pork, corn, wheat and soybeans into China. The new equilibrium reflects the scenario. By comparing differences between the baseline and the scenario, we estimate the impacts of the specific tariffs on US and global agricultural markets. Output from the CARD model is then used as input into the IMPLAN model and the GHG model.

2.2 | IMPLAN model

The input-output IMPLAN model is an inter-industrial accounting system that produces input-output accounts by region.[3] It is populated with data that are updated annually and is used to estimate the economic impacts of changes in regional production. Input-output models are price-static models that rely on economic characteristics of the recent past to project

[3]For more details about the IMPLAN model and its application, see https://www.implan.com/application/.

near-term outcomes. Modifications are made to the national model to more adequately reflect the crop and animal production sectors measured for this analysis.

The IMPLAN model translates percentage changes in the quantities produced of the targeted commodities into standard economic impact summaries. The current national model is modified to explicitly include the agricultural and manufactured commodities specified in this study. Percentage changes in output in the relevant commodity sectors generated by the CARD model are used to shock the model and produce multiplied-through impacts in terms of the direct effects on a particular industry or commodity, the indirect effects upon supply chains, and the induced effects caused by changes in labour income and household consumption. Direct activity describes the specific industry's total output, all payments that it makes to value added and to intermediate inputs, as well as the number of jobs in that industry. Indirect activity is the supply-chain-based economic activity that is stimulated by the purchase of goods and services as inputs into the direct sector. As the direct sector's demand for goods and services changes, so does the amount of economic activity in its supply chain. Induced activity occurs when jobholders in the direct and indirect sectors convert their labour incomes into household spending, thus stimulating another set of business activities in the economy.

The total economic effect is the sum of the direct, indirect and induced activities. As indicated in Figure 1, output from the CARD model is used in the calculations performed by IMPLAN. Changes in production quantities of the directly affected commodities are fed to the IMPLAN model to calculate the resulting direct, indirect and induced effects. These effects are reported in terms of industrial output changes, value added changes, labour income consequences and job impacts. Note that the IMPLAN model has been modified to avoid double counting of effects that are already accounted for in the CARD model. For example, the decline in domestic feed demand for corn and soybean meal due to reduction in pork production is factored into the CARD model scenario and the input-output model is modified to avoid double counting corn and soybean production losses due to decreased pork demand. Total output represents the value of goods and services that are produced by an industrial sector over the course of a year or, in the case of a shock to the model, changes in final demand in an industry's output. In producing total output, payments are made to labour income, to investors, and to governments in the form of indirect tax payments. These payments constitute value added, which are analogous to gross domestic product (GDP). Labour income is a subset of value added, and it reflects the value of wages and salaries received by workers as well as the value of employer-provided (or mandated) benefits. Jobs are the number of full and part-time jobs in the economy; the model does not report out full-time equivalencies. The IMPLAN model has been used in peer-reviewed publications to evaluate a wide range of economic activity related to employment, productivity or forestry (Douglas & Harpman, 1995; Giesecke, 2011; Parajuli et al., 2018).

2.3 | GHG Model model

The GHG Model model is an add-on to the CARD model to evaluate carbon emissions associated with policy changes. The need to take carbon emissions from agricultural production into account emerged from the discussions about life-cycle emissions from biofuels (Searchinger et al., 2008). To evaluate land use, we differentiate between cropland, forest, pasture and barren land. The carbon coefficients are calculated using the methodology and data by Gibbs (2006) for mean and maximum coefficients. The same method is used to obtain minimum carbon coefficients based on the data by Olson et al., (1985). The estimation of carbon coefficients typically involves uncertainty, which is reflected in our analysis by providing upper and lower bounds in addition to the mean values. Given the spatial crop distribution presented in Monfreda et al., (2008), the carbon coefficients in native vegetation, and the distribution of

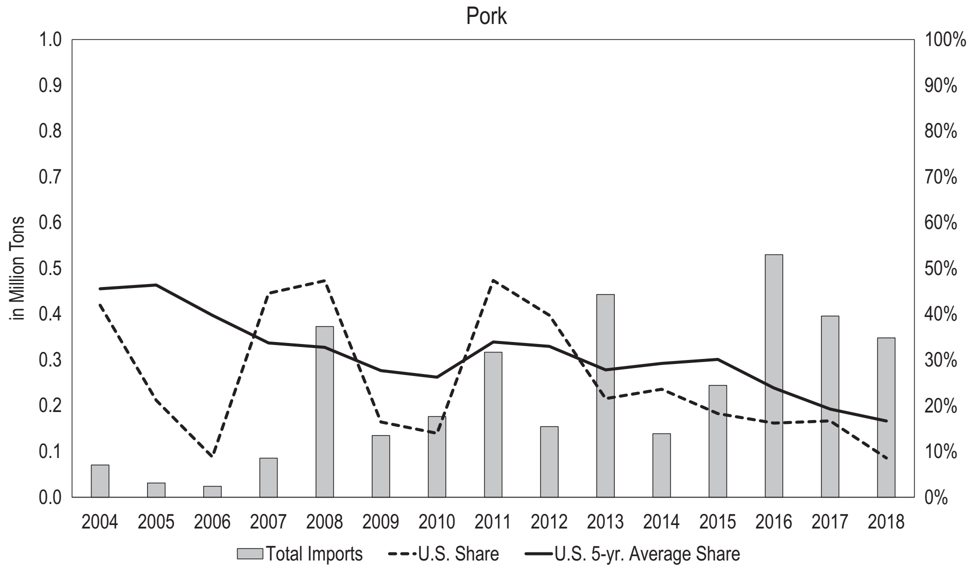

FIGURE 2 Total imports of pork to China as well as the share of US imports. The share is subdivided into annual and 5-year average share

various land types, we determine how a change in the crop area due to changes in trade flows affects carbon emissions from land use change.

2.4 | Scenario assumptions: Retaliatory tariffs by China

The scenario reflects the imposition of retaliatory tariffs by China on a set of US-produced agricultural commodities, that is, 25% tariffs on corn, soybeans and wheat, and 50% tariffs on pork. To implement the scenario, we first project the US market share in China based on the average of the last 5 years of the historical shares (before the initiation of the conflict, i.e., 2013–2017) of the imports to China from the United States for the commodities of interest (Figures 2, 3, and 4).[4] To implement the scenario, the baseline shares for US imports to China by commodity are exogenously reduced based on the magnitude of the retaliatory tariffs. These reduced shares are maintained throughout the projection period (Table 1). All the results obtained are conditional on observed reductions in the share of US imports to China that are consistent with the levels assumed here.

3 | RESULTS

We present three types of results in this section: (1) the US and global impacts of the retaliatory tariffs on agricultural markets, (2) the national economic impacts on the agricultural and industrial sectors using the IMPLAN model, and (3) the changes in GHG emissions from land use change associated with the retaliatory tariffs. In terms of the US and global impacts, we use the CARD model and focus mainly on the commodities directly affected by the tariffs. For the impacts on agriculture, results are expressed in percentage change relative to the baseline

[4]Only corn and wheat are side by side because the scales on the y-axis are identical for both commodities.

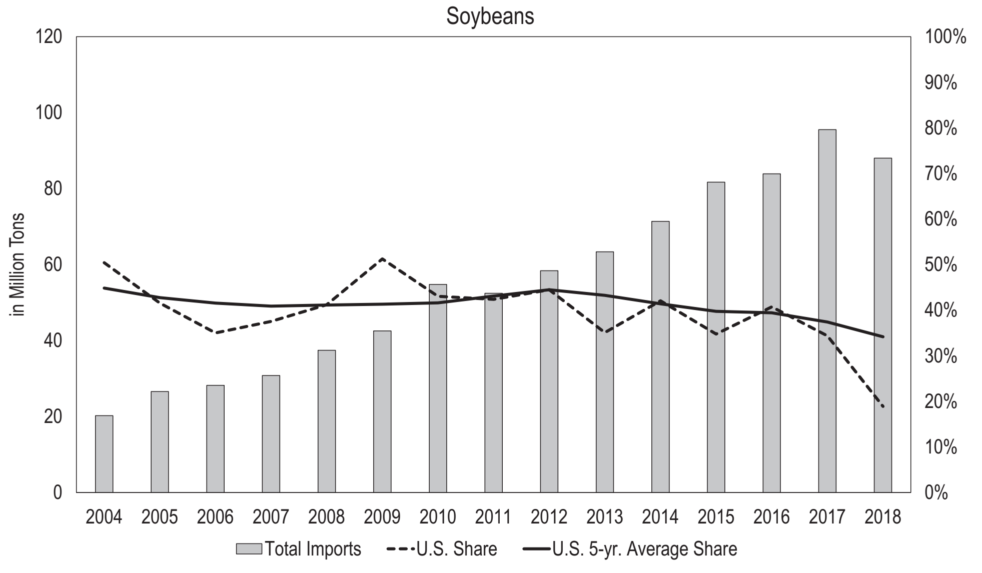

FIGURE 3 Total imports of soybeans to China as well as the share of US imports. The share is subdivided into annual and 5-year average share

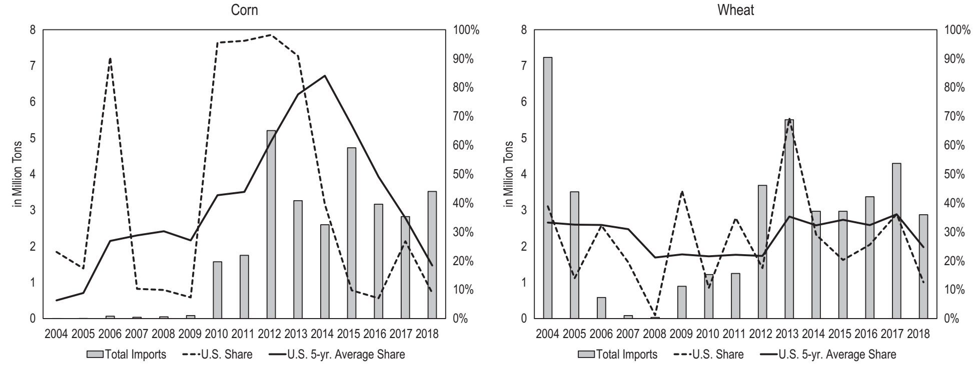

FIGURE 4 Total imports of corn and wheat to China as well as the share of US imports. The share is subdivided into annual and 5-year average share

TABLE 1 Baseline and scenario shares of US imports to China by commodity for the projection period

Commodity	Pork	Soybeans	Corn	Wheat
Baseline	19%	37%	35%	36%
Scenario	0%	15%	10%	10%

as the average of either the first 3 years (short term) or the last year (long term) of the projection period.[5] The short term represents the impact at the beginning of the projection period when the tariffs are first introduced. Because of the trajectories associated with the economic

[5]Although the results are discussed in terms of the short term (the average of the first 3 years of the 10-year projection period) and the long term (the last year), the tables also include results for the average of the 10-year projection period.

variables in the model (e.g., population and income changes over time), we also define the long-term effects to take into account the evolving nature of the macroeconomic environment. The long term also represents the effect of the tariffs as damages accumulate beyond the projection period given a long-lived trade conflict. The results are presented as an average of the first 3 years of the projection to reduce distortions that may be caused by initial fluctuations when the shock is introduced. The projections become smoother in the outer years of the period considered, and thus only the last year is presented.

In terms of the impacts on the national economy, we utilise the IMPLAN model to measure the total, direct, indirect and induced gains or losses to the US economy based on quantity changes in production of pork, soybeans, corn and wheat in the retaliatory tariff scenario relative to the baseline. The model is based on the initial output of each commodity to the US economy for 2017. The deviations between the baseline and the scenario production for each category are then modelled as proportionate change values relative to those respective anchor amounts had they occurred in 2017. All financial values in the following analyses are expressed in real 2017 dollars. Similar to the agricultural model, we differentiate between short and long term.

For the national economic impacts, results reflect the average annual deviations from the baseline under the scenario for the short term and the long term, one for each commodity and one that combines all four commodities into a group total (Table 4). The modelling does not factor in offsets, alternative uses of farmland, or other countervailing factors. The results also present changes in the number of jobs in the US economy as a result of the retaliatory tariffs considered.

Modifications in the areas allocated to crop production and on the livestock numbers in different countries can also lead to changes in the levels of GHG emissions that can be traced to the ongoing trade conflict. In this line, we use the GHG Model model to measure changes in the emissions of different countries as crops and livestock production reallocates across the globe.

3.1 | Impact on US production and markets of targeted commodities

As previously indicated, the CARD model solves for equilibrium in the world market for a wide range of commodities, but for the purposes of this study, we only report a subset of results on the commodities targeted by the retaliatory tariffs. However, some information on the reallocation of production in different parts of the world can be gleaned from the figures and discussions on changes in GHG emissions that result from the tariffs (as outlined in the following sections).

3.1.1 | Impact on US pork and other meat markets

In the case of pork, two distinct impacts can be observed when comparing the short versus the long term (Table 2). The price of pork, in particular the producer price (Barrows/Gilts), and the consumption variables (especially trade) adjust immediately after the introduction of the retaliatory tariffs. However, inertias of the processes and long cycles in animal production result in lower responses on the production side. Thus, in the short term (represented by the average of the first 3 years), producers see lower prices without being able to quickly adjust production, leading to economic losses. As economic losses accumulate, some producers leave the sector while others reduce production, which results in a gradual decrease in production over the projection period.

TABLE 2 Percentage change (scenario relative to baseline) in supply and utilisation for pork in the United States over the projection period

	Short term	10-year mean	Long term (Year 10)
Supply			
Imports	−3.98%	−3.82%	−4.68%
Production	−0.27%	−0.42%	−0.49%
Total	−0.38%	−0.52%	−0.61%
Disappearance			
Domestic use	0.85%	0.77%	0.81%
Exports	−4.32%	−4.45%	−4.70%
Total	−0.38%	−0.52%	−0.61%
Price			
Retail	−0.56%	−0.66%	−0.78%
Barrows/Gilts	−2.05%	−1.93%	−2.15%

Overall, US pork exports are reduced by 4.3%–4.7% over the projection period. The relatively constant decline in US pork exports over time in the scenario reflects the fact that pork exports do not increase significantly over the projection period in the baseline. With the complete loss of the Chinese market, the fall of US pork exports is slightly more pronounced in the outer years of the projection period than in the earlier years. The retaliatory tariffs on US pork from China reduce the global demand for pork. The lower demand leads to a decline in Barrows and Gilts (51%–52% live equivalent) pork prices by 2.1% for US producers in the short term and 2.2% in the long term (represented by the last year of the projection) (Table 2). The table shows that adjustments in prices and trade occur from the beginning of the projection and last until the end of the projection period.

The reduction in global demand for US pork is more pronounced than that of total demand as Chinese tariffs are specifically targeting pork of US origin. The pronounced impact on the global demand for US pork reflects a combination of two different effects. The tariff results in reduced total trade, that is, global trade destruction. There is also substitution of trade by China away from the United States and towards competing countries in the global markets, that is, trade diversion. As a result, countries such as Brazil, Canada and the European Union increase their participation in the international markets by increasing their net exports, taking the place of the United States (not shown).

In the case of the United States, lower prices encourage domestic use, which increases by 0.9% and 0.8% in the short and long term, respectively, because consumers can quickly adjust consumption patterns in response to changes in prices. The lower domestic price of pork also leads to a decline in production by 0.3% in the short term and 0.5% in the long term. These declines in production occur because the reduction in pork prices are only partially offset by lower feed costs as corn and other feed grain prices decline, also (in part) as a result of the retaliatory tariffs. Domestic disappearance of beef declines on average by 0.2% over the projection period despite the decrease in beef prices (0.2% in the short term and 0.1% in the long term). This comes from a substitution away from beef to the now relatively cheaper pork. Lower domestic beef demand allows for enhanced exports to the global market.

As mentioned previously, given the dynamics in animal production, farmers have a much more limited ability to adjust production in the first year than as time progresses explaining the smaller reduction in domestic production in the first year compared to the last year (as shown in Table 2). Over time production can also be increased in other countries given that total US production and exports decline. Either of these factors could explain a slightly increasing (over time) percentage reduction in US exports relative to the baseline.

The prices of pork and beef impact each other and are also affected by the prices of grains and oilseeds as these affect costs of production and the land devoted to pastures (and thus beef supplies) in countries such as Brazil. Pork and beef are substitutes in demand, thus lower US pork prices reduce domestic demand for beef and beef prices in the first few years of the projection. However, the reduction in the price for beef is relatively low in comparison to the price of pork, thus domestic beef consumption declines leading to expanding US beef exports, putting some downward pressure on international beef prices. While the US prices for pork, corn, wheat and soybeans (the commodities affected by the tariffs) decline, the representative world prices increase, leading to higher production in other agricultural exporting countries such as Brazil. A counteracting effect is that crops and beef production compete for land and especially for pasture-based beef production (the dominant system in Brazil). As crops outcompete cattle production for land, the new equilibrium that emerges has lower pasture area and lower beef production.

3.1.2 | Impact on US crops

For soybeans, Chinese retaliatory tariffs have a strong impact on US export levels, which fall by 29.0% and 31.2% relative to the baseline in the short and long term, respectively (Table 3). China's share of soybean import markets amounts to roughly 60%. The reduced Chinese and global demand for soybeans leads to a 15.8% reduction in the price received by US farmers in the long term. Notice that, in this case, prices of soybeans drop relative to the baseline in a significant way by 14.8% in the short term, mainly because of the immediate reduction in the export demand for US soybeans following the retaliatory tariffs.

Lower prices lead to a decline in soybean area harvested and yield, resulting in a reduction in soybean production by 6.9% and 8.3% in the short and long term, respectively. It should be noted, however, that the production declines are mitigated by reductions in returns of crops competing with soybeans for area (for example, corn and wheat, which also see lower prices due to the retaliatory tariffs). Given that soybeans are an annual crop, producers can reduce area harvested in the short term. The behaviour of the prices and harvested area of soybeans, both declining in the short term and then remaining at similar levels, indicates that the impacts in terms of lower revenues (as a result of the changes in both prices and production quantities) are felt even in the short term.

As reflected in Table 3, relatively cheaper soybeans encourage crushing and, thus, production of soybean meal and soybean oil. The United States shifts away from exporting soybeans to a scenario in which more soybeans are processed domestically and the production and exports of soybean meal and soybean oil increase. The expansion in exports of soybean meal (as a result of more production) is enhanced by the lower domestic use of soybean meal due to the reduced size of the pork sector after the tariffs are imposed. The capacity to shift to exports of co-products rather than soybean seeds embedded in the model might be behind this study's larger reduction in exports of oilseeds as compared to other studies (see Table 5 in the Discussion and Conclusions section).

Additional supplies and lower domestic demand (in the long term as the pork sector contracts) have the effect of lowering prices of soybean meal by 0.4% in the long term, relative to the baseline. Lower prices of soybean meal tend to reduce returns to crushing and the supply of soybean oil. However, as the demand for soybean oil remains strong, the price of this co-product tends to increase with the tariffs. Additionally, a reduction in the price of soybeans favours production of other oilseeds like canola and sunflower as these crops become relatively more profitable and better able to compete for land (even as they also suffer a price decline).

The increase in production of higher oil content crops like sunflower and canola (not shown) is a response to higher vegetable oil prices, and lower prices of soybean and other crops that compete for area such as wheat and corn. The relative magnitude of the changes of soybean oil to soybean meal is affected not only by supply considerations (in terms of what happens to soybean and other oilseeds with higher oil concentrations) but also by demand considerations (in terms of local and international demand for oil meals). Lower international supplies for soybean result in excess demand for vegetable oils and oil meals. However, the demand for oil meals is weakened by a smaller livestock sector. In this sense it is plausible to see prices of soybean oil that become higher relative to prices of soybean meal, even in the presence of expanded sunflower and canola production.

Corn is also affected by the tariff but in a muted way when compared to soybeans as a result of the tariff. The main reason is that China is a much larger destination for soybeans than for corn, and in particular for commodities of US origin. Thus, the impacts on US supply and utilisation for corn are in large part explained by spillover effects from the soybeans market.

The relatively large reduction in the price of soybeans lead to an increase in the area of corn (of about 1.2%–1.3%), despite a reduction in the price received by farmers of about 4% for this commodity both in the short and long term (Table 3). The expansion in production is only partially offset by the additional domestic demand encouraged by lower prices, and exports of corn increase by about 4% throughout the projection. The expansion in domestic corn demand in general and in demand for corn for feed in particular are in part muted by the contraction of the pork sector in the United States.

As with corn, China has a small proportion of the global import market for wheat, averaging roughly 2% between 2010 and 2017. The United States has a similar share in China's wheat market when compared to soybean and corn (Table 1). Because of these factors, the impacts on prices, production variables and trade are also relatively muted for wheat when compared to soybeans. Furthermore, and as in the case of corn, the changes in some of the variables of interest are strongly influenced and need to be explained with the movements in the market for soybeans. Similar to corn, wheat area is expected to increase despite the resulting lower prices for the commodity both in the short and long terms (Table 3). Although important in terms of economic activity and farm income, as mentioned above, the effects on wheat are lower when compared to the changes in soybeans, and soybean products. In the short term, there is a price reduction of close to 4% relative to the baseline, with the decline in price reaching about 5% in the long term. Despite the price reduction, US wheat area harvested is higher in the scenario by about 0.9% throughout the projected period, when compared to the baseline. The increase in area is due in part to a decline in the price of corn already presented, but more importantly by comparatively a larger decline in the prices of the crops that compete for land, such as soybeans, which impacts relative returns. Lower prices lead to an increase in all domestic uses. Differently from corn, the expansion in domestic use is larger than that of production, leading to a decline in wheat exports. This is also due in part to the lower export demand elasticity for wheat (relative to corn) used in this analysis (see footnote 3).

Note that the CARD model does not include any support given to farmers because of the tariffs. To mitigate the negative impact of the retaliatory tariffs on US farmers, the US government announced payments to agricultural producers under an initial $12 billion trade aid package in 2018 followed by another $16 billion in 2019. As part of the package was the Market Facilitation Program (MFP), which provided farmers with nearly $24 billion in direct payments (CRS, 2019b). Janzen and Hendricks (2020) find that, at least in the short run, the MFP payments tended to be larger than the estimated losses to farmers and also to favour some regions over others. The inclusion of this support would have dampened the effects of the retaliatory tariffs on the farming sector. However, the purpose of the study is to isolate the impact of the tariffs in order to assess the magnitude of the losses associated exclusively with the trade dispute. This approach may help inform policymakers in terms of the appropriate size of

TABLE 3 Percentage change (scenario relative to baseline) in supply and utilisation for corn, soybeans and wheat in the United States over the projection period

	Short term	10-year mean	Long term (Year 10)
Soybean supply			
Harvested area	−6.44%	−6.90%	−7.43%
Supply	−5.61%	−5.96%	−6.50%
Production	−6.88%	−7.44%	−8.31%
Soybean demand			
Domestic use	9.34%	9.14%	8.49%
Crush	9.94%	9.83%	9.25%
Seed and residual	1.88%	0.59%	−0.85%
Exports	−28.97%	−29.59%	−31.17%
Soybean price			
Farm price	−14.78%	−14.66%	−15.79%
Illinois processor price	−14.27%	−14.16%	−15.25%
Net returns	−20.93%	−20.84%	−22.92%
48% soymeal	0.44%	−0.12%	−0.44%
Corn supply			
Harvested area	1.33%	1.15%	1.18%
Supply	1.57%	1.56%	1.62%
Production	1.36%	1.16%	1.10%
Corn demand			
Domestic use	0.82%	0.73%	0.72%
Feed and residual	1.91%	1.76%	1.73%
Food and other	0.38%	0.41%	0.41%
High-fructose corn syrup	0.03%	0.04%	0.05%
Fuel alcohol	0.10%	0.08%	0.07%
Seed	0.96%	0.94%	0.96%
Exports	3.54%	4.24%	4.02%
Corn price			
Farm price	−3.65%	−3.89%	−4.19%
Net returns	−5.40%	−5.78%	−6.39%
Wheat supply			
Harvested area	0.93%	0.88%	0.87%
Supply	0.81%	0.94%	1.01%
Production	0.80%	0.73%	0.61%
Wheat demand			
Domestic use	0.79%	0.87%	1.08%
Feed and residual	3.66%	4.06%	5.66%
Food and other	0.30%	0.34%	0.39%
Seed	0.89%	0.83%	0.75%
Exports	−0.23%	−0.14%	−0.53%
Wheat price			
Farm price	−3.86%	−4.36%	−5.08%
Net returns	−6.47%	−7.34%	−8.73%

TABLE 4 Average annual deviation of the scenario from the baseline (in million dollars) of the relevant economic variables for pork, corn, soybeans, and wheat production change in the United States for the projection period

	Short term			Long term		
Impact type	**Labour income**	**Value added**	**Output**	**Labour income**	**Value added**	**Output**
Pork						
Direct	−29.7	−35.2	−60.7	−56.5	−67.0	−115.7
Indirect	−10.4	−16.8	−40.7	−19.7	−32.0	−77.5
Induced	−17.5	−30.6	−55.9	−33.3	−58.2	−106.4
Total	−57.5	−82.6	−157.3	−109.5	−157.2	−299.6
Soybeans						
Direct	−656.3	−835.0	−1,985.5	−983.4	−1,251.2	−2,975.2
Indirect	−537.3	−938.7	−1,807.8	−805.2	−1,406.6	−2,708.9
Induced	−555.2	−970.6	−1,774.8	−832.0	−1,454.4	−2,659.5
Total	−1,748.9	−2,744.3	−5,568.2	−2,620.6	−4,112.1	−8,343.6
Corn						
Direct	63.3	141.4	621.3	58.9	131.5	577.5
Indirect	211.5	388.9	761.3	196.6	361.5	707.6
Induced	129.8	226.8	414.8	120.6	210.8	385.5
Total	404.6	757.1	1,797.4	376.1	703.7	1,670.7
Wheat						
Direct	8.2	17.7	77.7	6.4	13.9	60.9
Indirect	27.3	48.6	95.0	21.4	38.1	74.4
Induced	16.8	28.4	51.9	13.1	22.2	40.7
Total	52.3	94.7	224.6	41.0	74.2	175.9
Combined: Pork, soybeans, corn and wheat						
Direct	−614.4	−711.0	−1,347.3	−974.6	−1,172.9	−2,452.5
Indirect	−308.9	−518.0	−992.2	−606.9	−1,039.0	−2,004.3
Induced	−426.2	−746.1	−1,364.1	−731.5	−1,279.6	−2,339.7
Total	−1,349.5	−1,975.0	−3,703.5	−2,313.0	−3,491.4	−6,796.6

Note: The column labelled 'Output' refers to gross output and includes intermediate inputs.

potential payments to farmers by commodity. Additionally, the analysis includes the spillover effects on other countries, which have long-term implications on US farmers that have to be taken into account.

Also critical to the model results is the value of β_i in Equation (7). If the value is small, it suggests US products and products from other exporters are poor substitutes in non-China markets, and the net result will be that the prices can diverge. If the value is very large, on the other hand, US and competitor exports are close substitutes, and prices will not diverge as much in the presence of a shock like that examined here.[6] To examine the sensitivity of our

[6]The central values of the β_i in Equation (7) were informed by the own price elasticities for US export demand for soybeans, corn and wheat estimated by Reimer et al., (2012) and for pork estimated by Kaiser (2012). These elasticities are −1.11, −0.90, −0.45 and −1.12 for corn, soybeans, wheat and pork, respectively.

TABLE 5 Percentage changes relative to the baseline of selected variables obtained by this and other recent studies as the result of tariffs imposed on different commodities and countries for different modelling frameworks

	Taheripour and Tyner (2018)	Zheng et al. (2018)	This study	
Type of model	**CGE**	**Partial equilibrium**		
Time frame	**Long-run**	**Short-run**	**Short-term**	**Long-term**
Percentage change relative to a baseline				
Soybean trade				
US exports	−24.1%/−34.2%		−29.2%	−32.8%
US exports to China	−47.7%/−90.6%	−34.2%	−60.5%	−60.3%
Total trade	−2.5%/−0.2%		−4.4%	−4.1%
Pork trade				
US exports			−4.3%	−4.6%
US exports to China		−83.3%	−100%*	100%*
US production				
Soybeans	−10.6%/−14.7%	−1.6%	−6.9%	−8.3%
Wheat	1.9%/2.9%		0.80%	0.61%
Corn			1.4%	1.1%
Grains	1.2%/1.7%			
Pork		−0.2%	−0.27%	−0.5%
Prices				
Soybeans	−4%/−5%	−3.9%	−14.8%	−15.8%
Wheat			−3.9%	−5.1%
Corn			−3.6%	−4.2%
Pork		−0.6%	−2.1%	−2.2%

Notes: Taheripour and Tyner (2018) assess tariffs on corn, soybeans, wheat and beef. The separate values in the table refer to the standard and elevated GTAP trade elasticities, respectively. Notethat the elevated trade elasticities are implemented for soybeans only. Zheng et al. (2018) analyse tariffs on cotton, soybeans, sorghum and pork.

*By design (Table 1), the US share of soybean imports of corn are 0%, thus precluding the entry of US pork into China.

results to the parameters, we have conducted a sensitivity analysis and varied the parameter by ±20%. The change in percentage points for those variables that have a change of at least 0.25% in one direction are presented in Figure 5.

3.2 | Impacts on the US national economy

This section presents the effects on some indicators of economic activity that result from the modifications of the markets presented in the previous section. The reduction in pork production has effects not only on commodity markets but also on employment and economic activity (Table 4). In the short term, direct, farm-level output decreases by $61 million, resulting in about 624 lost jobs (including farmers), reflecting lost earnings totalling nearly $30 million in labour income (Table 4). This reduction in pork production indirectly suppresses $41 million of output at the national level supporting 185 jobs with annual earnings totalling $10.0 million. When considering the direct and indirect sector jobholders converting their labour incomes into household spending, an additional output contraction of $56 million is estimated,

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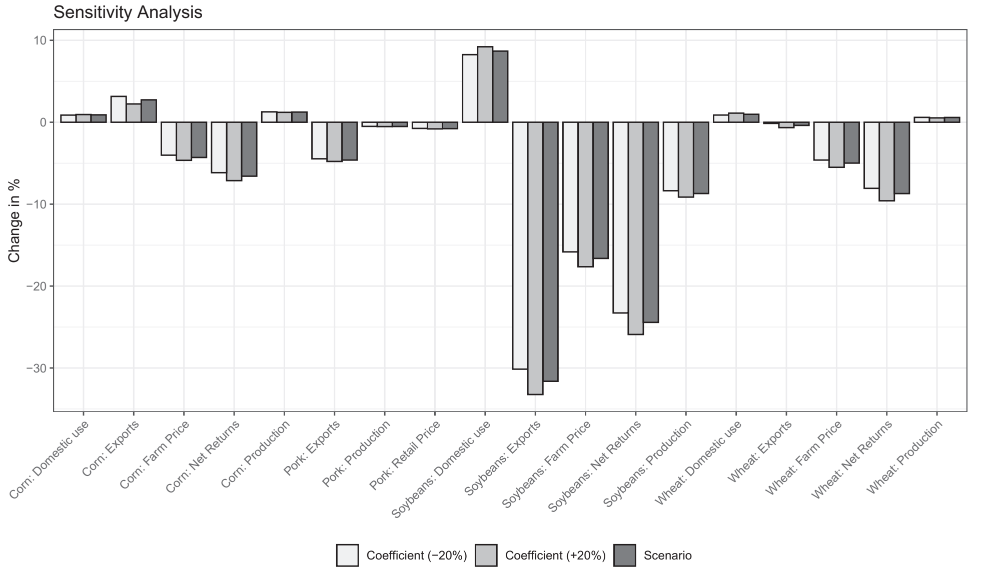

FIGURE 5 Sensitivity analysis of varying β_i in Equation (7) by ±20%. Only coefficients that change by more than ±0.25% are depicted

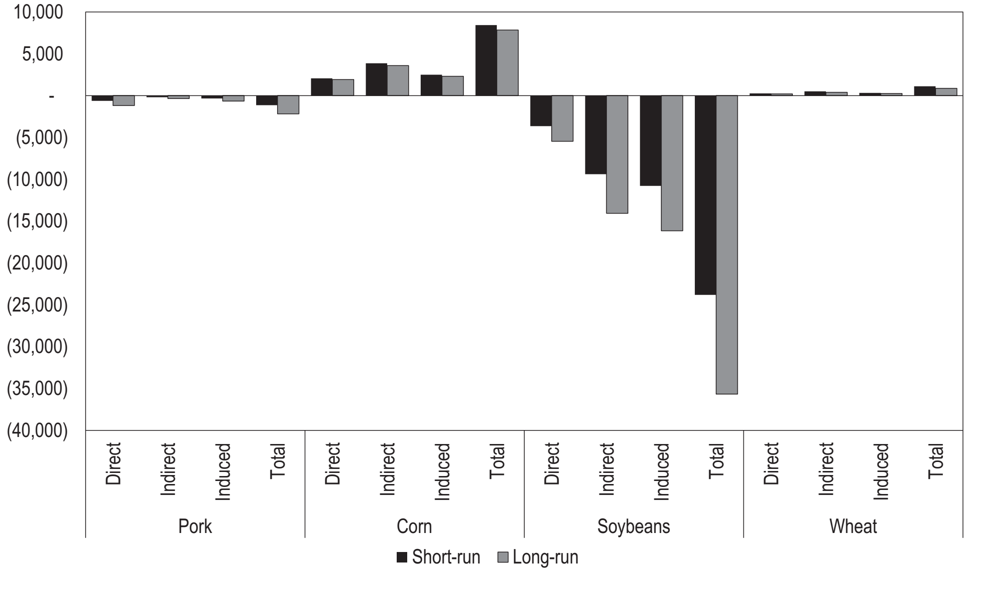

FIGURE 6 Average annual deviation of the scenario from the baseline of employment for the various sectors

resulting in the elimination of 337 jobs and a loss in labour income of nearly $18 million. Combined, the loss in pork production would cost the economy $157 million in total national output and $83 million in value added, of which $58 million would be labour income of 1146 jobholders (Figure 6).

In the long term, a reduction in pork production results in $300 million lower total US industrial output and $157 million in reduced value added, of which $110 million is the labour income lost due to 2182 fewer jobholders. Although it is tempting to focus on the largest value in the table, the output amount, we properly measure gains or losses to the economy using the value-added amount, given that value added is analogous to GDP. Accordingly, we conclude that, in the last year of the projection period, the annual average cost nationally in the scenario relative to the baseline is $157 million in value added (or GDP) for pork production.

Scenario-driven soybean production impacts on the economy are also presented in Table 4. In the short term, production declines result in a $5568 million fall in total output nationally and $2744 million decrease in total value added, of which $1749 million is a loss of labour income from 23,804 lost jobs annually, over the period measured (Table 4). After all direct effects multiply through, production declines result in $8344 million lower total output nationally and $4112 million in reduced total value added, of which $2621 million would be lost labour income from 35,669 fewer jobs annually (Figure 6).

For the case of corn, as indicated in the previous section, the impacts on area and production are positive for both the short and long term. In terms of short-term national economic impacts related to changes in the corn sector, Table 4 results show an increase of $621 million in direct farm output, resulting in 2072 more farm-level jobs and about $63 million in increased labour income. After considering all direct, indirect and induced effects, short-term corn production gains boost the economy by $1797 million in total national output and $757 million in value added. Labour income increases by $405 million and the total number of jobs rises by 8438. For the long term, the corn sector total national output gains average 1671 million with nearly $704 million in value added, including $376 million in labour income and 1671 jobs.

In the short term, the increase in wheat production results in an additional $225 million in total output nationally and $95 million in value added. Labour income increases by $52 million from 1121 additional jobs. There are also ongoing long-term gains in wheat production. Total output grows by $176 million and value added increases by $74 million, of which almost $41 million would be labour income from 878 jobholders.

When combining the national economic impacts for pork, soybeans, corn and wheat, the short-term average annual scenario change over the baseline results in $3.7 billion in reduced national total output and $2.0 billion in lower value added, including $1.4 million in lost labour income attributable to 15,391 fewer jobs, on average, over the projection period. In the long term, national output declines by $6.8 billion while value added falls by $3.5 billion, of which $2.3 billion is lost labour income arising from 29,129 fewer jobs.

3.3 | Greenhouse gas emissions

The GHG emissions are based on the last year of the projection period. The emissions are calculated for minimum, mean and maximum carbon coefficients (Carriquiry et al., 2020; Dumortier et al., 2012, 2020). The pasture area in the CARD model is explicitly modelled for Brazil but uses an ad-hoc approach based on stocking rates, livestock size and pasture area for other countries (Dumortier et al., 2012). The GHG model also incorporates the change in pasture area associated with changes in herd size.

Basing carbon emissions only from land use on cropland, we find an increase in emissions across all assumed carbon coefficients of 19.7–266.0 teragrams (Tg) of CO_2-equivalent (CO_2-e). This suggests that the trade conflict between the United States and China increases GHG emissions if solely cropland is considered.

If pasture is also considered, the effects on GHG emissions are reversed. Emission for minimum carbon coefficients are lower by 83.1 Tg CO_2-e in the scenario compared to the baseline. Assuming mean and maximum coefficients, the GHG emissions are also lower by 77.3 Tg CO_2-e and 67.5 Tg CO_2-e, respectively. The reduction in emissions when pasture is considered suggests a contraction in pasture area, which compensates for the increase in crop area. This is especially notable in Brazil. Due to the decrease in feed cost, beef cow and dairy cow herds increase in the United States. The same effect is observed in China and the EU. Australia, Japan and Mexico all see an increase in dairy cows. All other cattle categories decrease. This reduces the global pasture requirements. Thus, when we calculate the GHG emissions from cropland and pasture combined at the global level, the emissions are lower. Similar offsetting GHG emission effects are highlighted by Richards et al., (2020). The reason for this is that the land use reduction in Canada and the United States is sufficient to outweigh the increase in land use in carbon-rich environments such as Brazil. The soybean area in Brazil increases by 732,000 hectares, whereas the area for all other crops in Brazil declines. Livestock in Brazil contracts because it is less cost-competitive when compared to US livestock under the trade tariffs. The reduction in crop area harvested in Canada and the United States and the associated biomass and soil carbon content is enough to compensate for higher emissions in Brazil and in other countries (Figure 7).

Early estimates by Fuchs et al., (2019) suggested an increase in Brazilian soybean area by up to 12.9 million ha assuming that the reduction in US exports to China is completely compensated by Brazil. This number is reduced to 5.7 million ha in their analysis if the shortfall of US exports is compensated proportionally by all other soybean producers. Their analysis is not based on using a (partial or general equilibrium) trade model but based on back-of-the-envelope calculations using FAO data. Our calculations report an increase in Brazilian soybean area by 1.3 million ha. Similar to the results by Richards et al., (2020), the importance of pasture in Brazil is highlighted by our results. Using the GTAP model, Richards et al. (2020) find an increase in Brazilian soybean area by 4 million ha with most of that area coming out of existing crop and pastureland. This is reflected in our results as well, which show a large increase in GHG emissions if only cropland is considered but because pasture area declines in Brazil as a result of the trade conflict, overall emissions are smaller when both cropland and pasture are taken into account.

Verburg et al., (2009) find 6% higher GHG emissions under trade liberalisation, which is different from our results because we find a potential increase in emissions under trade restrictions. West et al., (2010) illustrate the trade-off between carbon content and crop production at

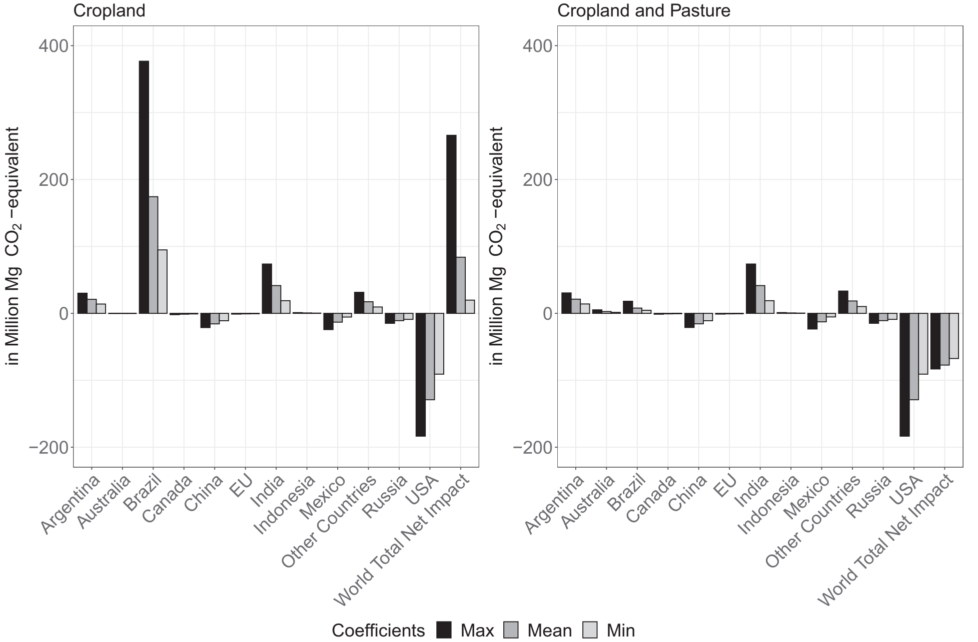

FIGURE 7 Difference in emissions from land use change between the baseline and the scenario

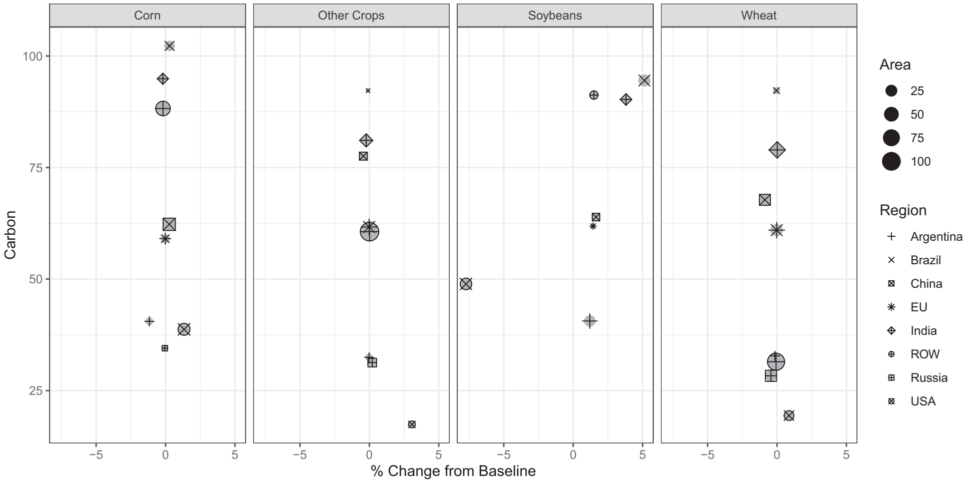

FIGURE 8 Mean carbon content in native vegetation and change in crop area between baseline and scenario in the CARD model. The change in crop area represents the percentage change in the scenario compared to the baseline

the global scale. Dumortier et al., (2019) provide the average carbon content of native vegetation in various CARD regions. For our analysis, we illustrate the changes in crop area for corn, soybeans, wheat and other crops based on carbon content in native vegetation and crop area (Figure 8). He et al., (2019) find that the trade diversion of soybeans away from US exports increases environmental costs in the short run but could decrease environmental costs in the long run because Brazil and Argentina have a comparative advantage.[7] The reason for higher costs in the short run are longer trade routes from South America to China and a surplus of soybeans in the United States with farmers not adjusting cropping systems. In addition, Chinese soybean demand increases area in both Brazil and China in the short run. They find that in the long run, soybean trade would be lower, which results in positive environmental aspects.

4 | DISCUSSION AND CONCLUSIONS

This study examines the impact of retaliatory tariffs imposed by China on pork, corn, soybeans and wheat, using the CARD model in conjunction with the input-output IMPLAN model. We also evaluate the GHG emissions triggered by land use change from the trade conflict using a GHG model. We show that retaliatory tariffs result in trade diversion as imports to China are diverted from the United States to other suppliers, which reduces the US market share in China's imports. The tariffs also result in trade destruction as the aggregate level of trade is reduced. Domestic prices of the impacted agricultural commodities increase in China, which reduces consumption and growth in production leading to trade destruction. The impact of the Chinese retaliatory tariffs has welfare-reducing implications in the US agricultural sector. Agriculture-related jobs and, consequently, labour income declines because of a significant reduction (over $6.8 billion) in national output in the long term. In terms of environmental consequences, GHG emissions increase when only cropland is taken into account but decrease

[7]The environmental indicators considered in their analysis are energy requirement, nonrenewable energy requirement, water footprint, arable land occupation, mineral fertiliser footprint, carbon footprint, nitrogen volatilisation, nitrogen leaching and runoff, phosphate loss and pesticide residue.

when pasture is included because global pasture requirements decrease with a contraction in the livestock sector in carbon-rich countries like Brazil.

Although the directional changes in the economic variables mentioned here could have been anticipated based on conceptual economic models, the magnitudes of the effects is an empirical question and its quantification requires the use of agricultural projection models as employed in this study. The full magnitude or consequences of the trade distortions on the affected commodities are still to be seen and have not fully unfolded.

A number of studies have looked at how tariffs and trade restrictions affect agricultural markets (Cheptea & Gaigné, 2019; Cipollina & Salvatici, 2020; Grant et al., 2019; Nti et al., 2019; Taheripour & Tyner, 2018; Westho et al., 2019a; Zheng et al., 2018). As expected from economic theory, the introduction of tariffs and other barriers to free trade have the impact of reducing imports (and thus exports) at a global level. We compare the results of this study with the results presented by Taheripour and Tyner (2018) (using a general equilibrium model) and Zheng et al., (2018) (using a partial equilibrium model) in Table 5. It is important to note that the results from these studies provide some insight on the range of impacts but they are not directly comparable as modelling frameworks differ between the studies and there are variations in terms of the targeted commodities and countries imposing tariffs. Additionally, these models tend to focus on either the long-run or short-run implications of tariffs. Specifically, general equilibrium strategies have to assume the policy is extended through time and assess the long-run implications without informing the short-run. This paper uses a partial equilibrium model to assess the impact of retaliatory tariffs on both bilateral trade and global markets as well as in both the short and long term.

Our study presents several highlights. First, bilateral trade restrictions have global implications both in terms of agriculture and the environment. Given the size and market shares of the United States and China, the retaliatory tariffs by China on US agricultural commodities directly affect land allocations within the countries in question but also indirectly across other major exporting (and importing) countries. Additionally, trade barriers do not only affect the commodities directly affected by the restrictions. For example, Cheptea and Gaigné (2019) investigate the impact of the retaliatory ban on imported food products from the EU and the United States by Russia in 2014. They find that, in addition to EU exports of banned products to Russia declining by 80%, EU exports of non-banned products decline by 20% and the ban has negative welfare implications on Russian consumers.[8] Thus, bilateral trade restrictions have far-reaching consequences that affect people well beyond the countries engaged in the dispute and beyond the restricted commodities. Models with global coverage, such as the CARD model, are needed to understand the full ramifications of the trade dispute. Second, producers across the agricultural sector are directly and indirectly impacted by the trade dispute. Generally speaking, the profit margins of farmers whose products are directly affected by the retaliatory tariffs shrink. However, profit margins of other agricultural producers not directly impacted by the tariffs are also affected given the linkages between the different commodity markets. This has government support policy implications in terms of who to compensate and by how much. Third, this study also sheds light on the direct and indirect economic impacts on agriculture in terms of farm labour and farm income. Although beyond the scope of this study, farmers have also seen higher production costs because of tariffs imposed on steel and aluminum, which have increased equipment costs. In October 2019, the American Farm Bureau Federation estimated that Chapter 12 farm bankruptcies increased by 24% relative to the previous year.[9] If these national impacts are long term, this may lead to structural

[8]The depreciation of the Russian currency and a drop in oil prices during the same time as the ban also contributed to the loss in EU exports and consumer purchasing power in Russia.

[9]Farm Bureau, 'Farm Bankruptcies Rise Again: Chapter 12 Filings Increase 24% Compared to Year-Ago Levels', 30 October 2019, https://www.fb.org/market-intel/farm-bankruptcies-rise-again.

changes in the agricultural sector. Fourth, spillover effects are also evident in the emissions results. Our study shows that, for more insight on the environmental impacts, it is important to calculate GHG and carbon emissions when taking into account both changes in land use across countries as well as global changes in cropland and pasture. Otherwise, emissions will tend to be overstated, which also have implications when determining policies to mitigate GHG emissions (Richards et al., 2020). It is also worth noting that focusing on GHG emissions as an environmental impact of the tariffs, this study may be missing changes in other ecosystem services that occurs as land use changes occur in different parts of the world. In this sense, and as an example, biodiversity losses may occur in regions where land is brought into agricultural production (e.g., resulting in deforestation), while they may increase in places in which land becomes less intensively used (Fuchs et al., 2019). This study is silent in this regard, but poses the question for further research.

This study, and the strategy followed, allows us to isolate the expected changes in US and global commodity markets attributable to the trade conflict. Nonetheless, we are aware that other events that developed and evolved after the beginning of the conflict are also strongly influencing these markets. These events interact in important ways, reinforcing and/or offsetting part of the impacts quantified in this study. The outbreaks of African Swine Fever (AFS) in Asia and Europe, the global COVID-19 pandemic, and the more recent trade agreement between the United States and China (currently under Phase 1) are worth noting but are hardly the only events affecting global commodity markets. In this line the outbreak of AFS could have been expected to provide significant opportunities for US pork producers to expand output and exports. However, work has shown that most of these potential opportunities were eliminated by the ongoing trade conflict (Carriquiry et al., 2019). COVID-19 also has implications for food consumption (and consumption patterns at home or away from home), that are under active research, and have implications for global commodity markets. Phase 1 of the US–China trade agreement could also be offsetting some of the impacts quantified here. Although important, and currently an active area of research, the effects of these events on domestic and global commodity markets are beyond the scope of this study.

Finally, while the US government has spent billions of dollars in aid to farmers to dampen the negative effects of the retaliatory tariffs, we offer three overarching questions. Given that a significant proportion of farm income is currently from federal programmes, what are the long-term implications of the reliance of the farming sector on government support if the trade dispute is not resolved?[10] Will the United States be able to regain its lost shares in the global agricultural market as Brazil and other net exporting countries position themselves to fulfil China's growing import demand? What are the economy-wide implications of the trade dispute given that the retaliatory tariffs are not limited to agricultural commodities or to just China? Even though we isolate the retaliatory tariffs implemented by China only on a few (major) agricultural commodities, this study contends that the implications may be far-reaching, long-term and perhaps irreversible.

The nature of any economic model is that there are limitations. Our research question focuses on the impact of the retaliatory tariffs on the agricultural sector with implications on some economic indicators for the United States. There are many interacting agricultural markets covered in the CARD model. The economic activities that are related to changes in the agricultural market are captured with the IMPLAN model, which considers the changes in the size of the markets in the agricultural sector. We are interested in the labour and income effects of the agricultural markets and this is captured by the IMPLAN model. We are not looking at the effect beyond the agricultural markets. Those limitations can be overcome by using other models with different sets of limitations.

[10]USDA Economic Research Service Farm Sector Income and Finances, 'Highlights from the November 2019 Farm Income Forecast', 27 November 2019, https://www.ers.usda.gov/topics/farm-economy/farm-sector-income-finances/highlights-from-the-farm-income-forecast/.

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ACKNOWLEDGEMENTS

We are thankful to two anonymous reviewers who provided valuable comments to the manuscript as well as Dr. David Harvey (handling editor). The initial idea for this paper stems from a project funded by the Corn Refiners Association.

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How to cite this article: Elobeid A, Carriquiry M, Dumortier J, Swenson D, J. Hayes D. China-U.S. trade dispute and its impact on global agricultural markets, the U.S. economy, and greenhouse gas emissions. *J Agric Econ*. 2021;72:647–672. https://doi.org/10.1111/1477-9552.12430